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A. Performance of all models on $a + b =$

We test three models, GPT-J, Pythia-6.9B, and Llama3.1-8B on the task $a + b =$. At first, we attempted to prompt each model with just $a+b =$, but we achieved significantly better results by including additional instructions in the prompt. After non-exhaustive testing, we used the prompts listed in Table 2 to test each model for all 10000 addition prompts (for $a, b \in [0, 99]$). We plot a heatmap of the accuracy of the model by a and b in Fig. 11. All three models are able to competently complete the task, with Llama3.1-8B achieving an impressive 98% accuracy. However, in the main paper we focus on analyzing GPT-J because it employs simple MLPs that are easier to interpret. We note that all three models struggle with problems with larger values of a and b.

B. Additional Results on the Structure of Numbers

Our Fourier decomposition results in Section 4.1 are sensitive to the number of a values analyzed. In particular, we find that the $T = 2$ Fourier component is not identified when analyzing h^0_{361}, but is identified when analyzing h^0_{360} (Fig. 12). While we consider this sensitivity to sample size to be a limitation of our Fourier analysis, we note that the Fourier analysis is itself preliminary. We find that the $T = 2$ Fourier feature is causally relevant when fitting the residual stream in Section 4.4. Additionally, in later sections we find that neurons often read from the helix using the $T = 2$ Fourier feature, indicating its use downstream (Fig. 34). Thus, we identify $T = 2$ as an important Fourier feature.

We compare the residual stream of GPT-J after layer 0 on the inputted integers $a_1, a_2 \in [0, 99]$ using Euclidean distance and cosine similarity in Fig. 13. We visually note periodicity in the representations, with a striking period of 10. To analyze the similarity between representations further, we calculate the Euclidean distance between a and $a + \delta n$ for all values of δn. In Fig. 14, we see that representations continue to get more distant from each other for $a \in [0, 99]$ as δn grows, albeit sublinearly. This provides evidence that LLMs represent numbers with more than just periodic features. When a is restricted to $a \in [0, 9]$, we observe a linear relationship in δn, implying some local linearity. The first principal component of the numbers $a \in [0, 360]$ (shown in Fig. 15) is also linear with a discontinuity at $a = 100$. Thus, our focus on two digit addition is justified, as three-digit integers seem to be represented in a different space.

C. Additional Helix Fitting Results

C.1. Helix Properties

For the input of layer 0 of GPT-J, we plot the magnitude of the helical fit's Fourier features. We do so by taking the magnitude of columns of C in Eq. 2. We find that these features roughly increase in magnitude as period increases, which matches the ordering in the Fourier decomposition presented in Fig. 2.

Additionally, we visualize the cosine similarity matrix between columns of C, which represents the similarity between helix components. In Fig. 17, we observe that components are mostly orthogonal, as expected. A notable exception is the similarity between the $T = 100$ sin component and the linear component.

To ensure that the helix represents a true feature manifold, we design a continuity experiment inspired by Olah & Jermyn (2024). We first fit all a that do not end with 3 with a $T = [100]$ helix. Then, we project $a = 3, 13, \ldots, 93$ into that helical space in Fig. 18. We find that each point is projected roughly where we expect it to be. For example, 93 is projected between 89 and 95. We take this as evidence that our helices represent a true nonlinear manifold.

C.2. Additional Causal Experiments for Helix Fits

We first replicate our helix fitting activation patching results on Pythia-6.9B and Llama3.1-8B in Fig. 19.

To ensure the helix fits are not overfitting, we use a train-test split. We train the helix with 80% of a values and patch using the other 20% of a values (left of Fig. 20). We observe that the helix and circular fits still outperform the PCA baseline. We also randomize the order of a and find that the randomized helix is not causally relevant (middle of Fig. 20), suggesting that the helix functional form is not naturally over expressive. Finally, we demonstrate that our results hold when fitting the b token on $a + b$ with $\text{helix}(b)$ (right of Fig. 20). Note that when activation patching fits on the b token, we use clean/corrupted prompt pairs of the form $(a + b', a + b)$, in contrast to the $(a' + b, a + b)$ pairs we used for the a token.

We perform an ablation experiment by ablating the columns of $C^\dagger$ from each h^l_a. In Fig. 21, we see that ablating the helix dimensions from the residual stream like this affects performance about as much as ablating the entire layer, providing additional causal evidence that the helix is necessary for addition. However, when we attempt to fit a with $\text{helix}(a)$ for other tasks in Fig. 22, we find that while the fit is effective, it sometimes underperforms PCA baselines. This suggests that while the helix is sufficient for addition, additional structure is required to capture the entirety of numerical representations. For a description of the prompts